

Prime Numbers

Prime numbers are numbers which have only 2 distinct factors i.e 1 and the number itself. Eg. 2,3,5,7,19 etc.

Armstrong Numbers

Armstrong numbers are numbers which have their sum of cube of individual digits equal to the number itself.

E.g $153 = 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$.

Practice Problems

Ques1. Write a program to check if a number is prime or not.

```
#include <iostream>
#include <cmath>
using namespace std;

int main() {

    int n;
    cin >> n;

    bool flag = 0;

    for(int i = 2; i <= sqrt(n); i++){

        if(n % i == 0){
            cout << "Non-prime" << endl;
            flag = 1;
            break;
        }
    }

    if(flag == 0){
        cout << "prime" << endl;
    }

    return 0;
}
```

Ques2. Write a program to print all the digits of a positive decimal number from right to left

```
#include <iostream>
using namespace std;
int main()
{
    int n;
    cin >> n;
    while (n > 0)
    {
        int rem = n % 10;
        cout << rem << " \n ";
        n = n / 10;
    }
    return 0;
}
```

Ques3. Write a program to reverse a number

E.g Given a number 1879 we need to convert it to 9781.

100020 will be converted to 20001 (Note: We need to remove the trailing zeroes).

```
#include <iostream>
using namespace std;

int main() {

    int n;
    cin>>n;

    int reverse=0;
    while(n>0){
        int lastdigit= n%10;
        reverse = reverse*10 + lastdigit;
        n=n/10;
    }

    cout<<reverse<<endl;

    return 0;
}
```

Ques4. Write a program to check if a number is Armstrong number

```
#include <iostream>
#include<math.h>
using namespace std;
```

```
int main() {  
  
    int n;  
    cin>>n;  
  
    int sum=0;  
    int originaln=n;  
    while(n>0){  
        int lastdigit= n%10;  
        sum+= pow(lastdigit,3);  
        n=n/10;  
    }  
  
    if(sum==originaln){  
        cout<<"Armstrong number"<<endl;  
    }  
    else{  
        cout<<"not armstrong"<<endl;  
    }  
  
    return 0;  
}
```

Some more practice questions

- ✓ Ques5. Write a program to print the ASCII value of any given character.
- ✓ Ques6. Write a program to display all factors of a number.
- ✓ Ques7. Write a program to find the factorial of a given number.
- ✓ Ques8. Write a program to find whether a given character is a vowel or a consonant.

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